Software Engineer Lab

Unified Modeling Language: Use Case

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Getting Start

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Getting Start

1. Scoring Criterial

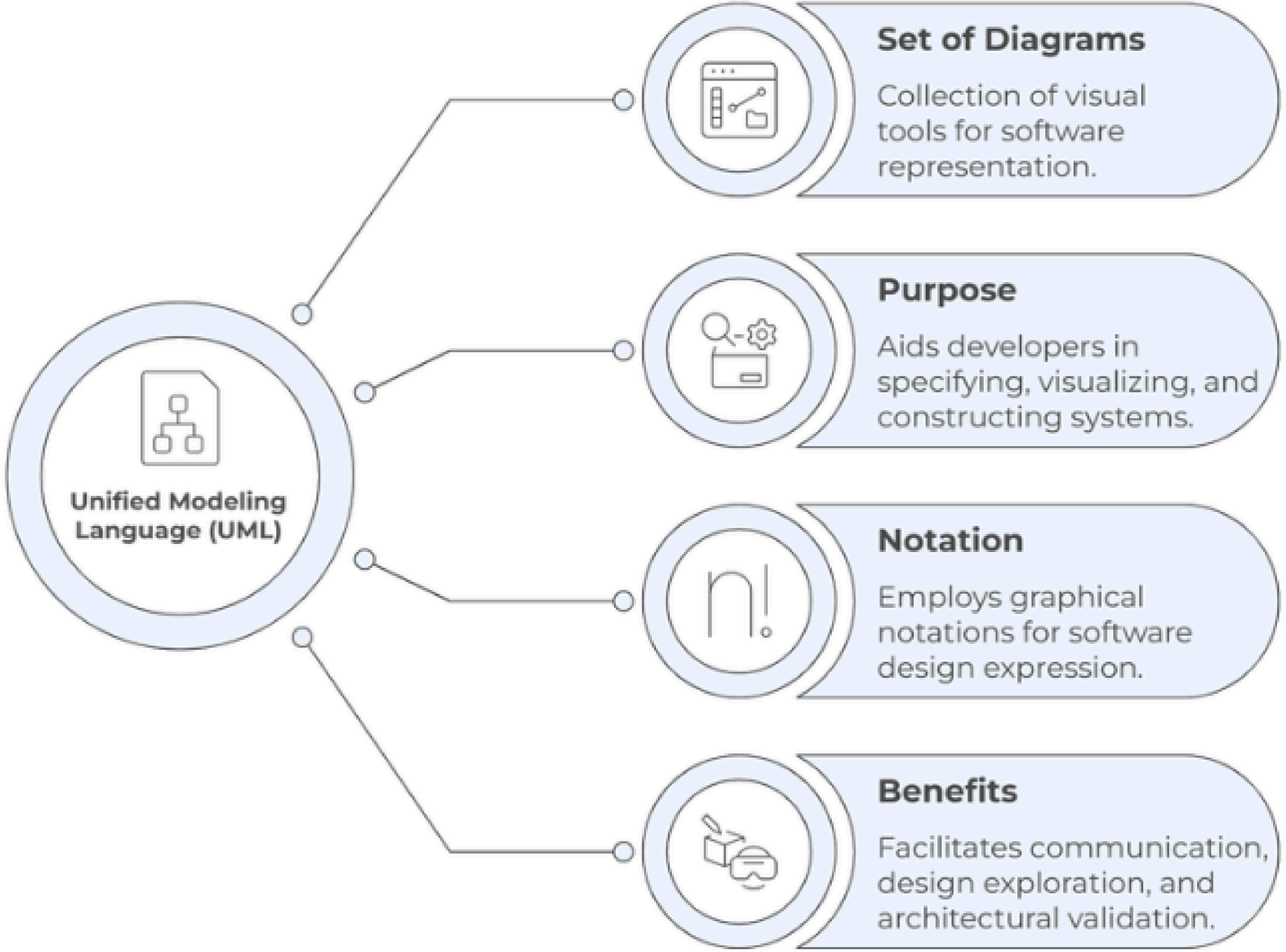
Section	Scoring
Attendance	10%
Homework/Assignment	15%
Quiz	15%
Final	30%
Project	30%

Getting Start

2. Scedule

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11
Use Case Diagram						
Activity Diagram						
Midterm + Project Introduction						
Class Diagram						
Sequence Diagram						
Final						
Presentation						

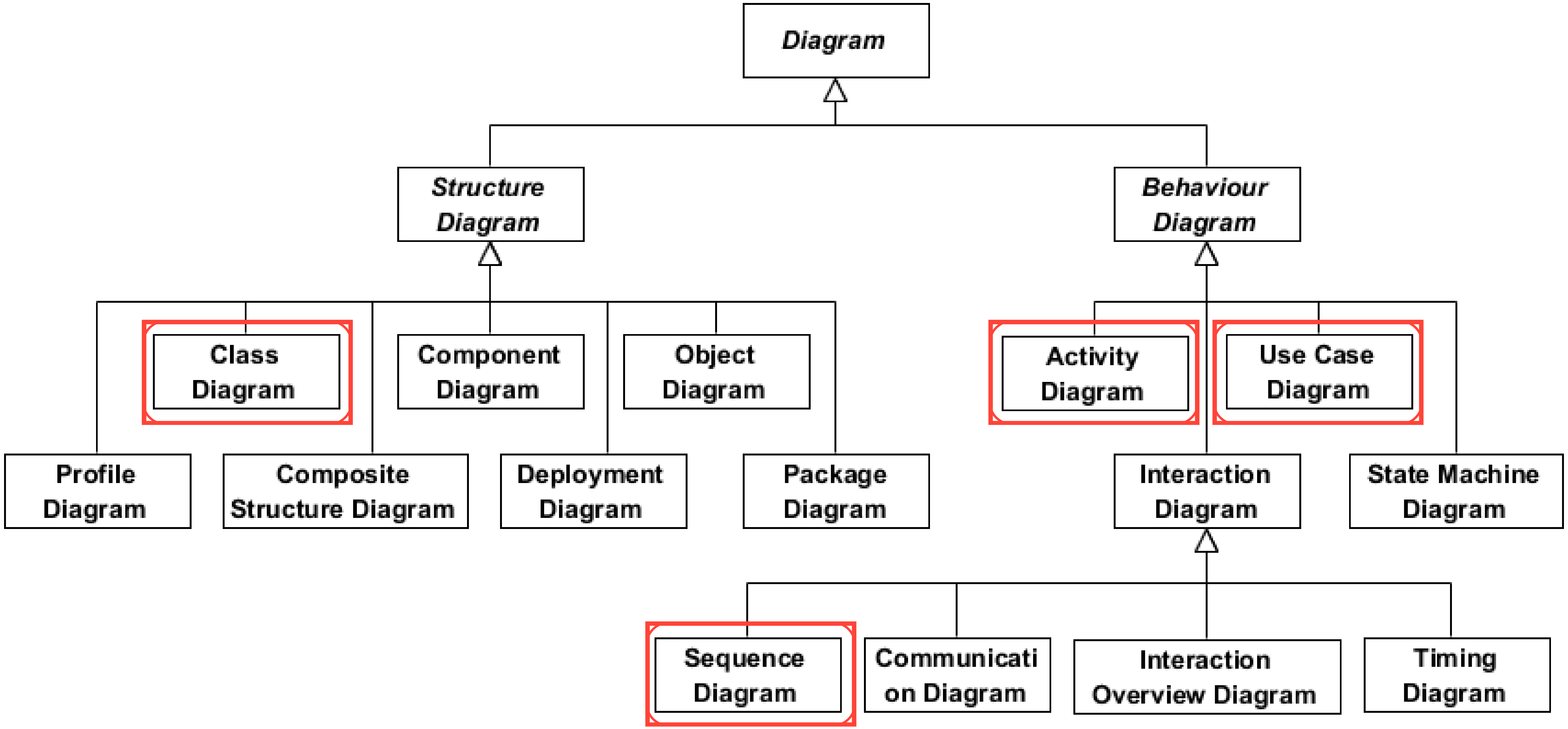
I. Introduction to UML



I. Introduction to UML (cont)

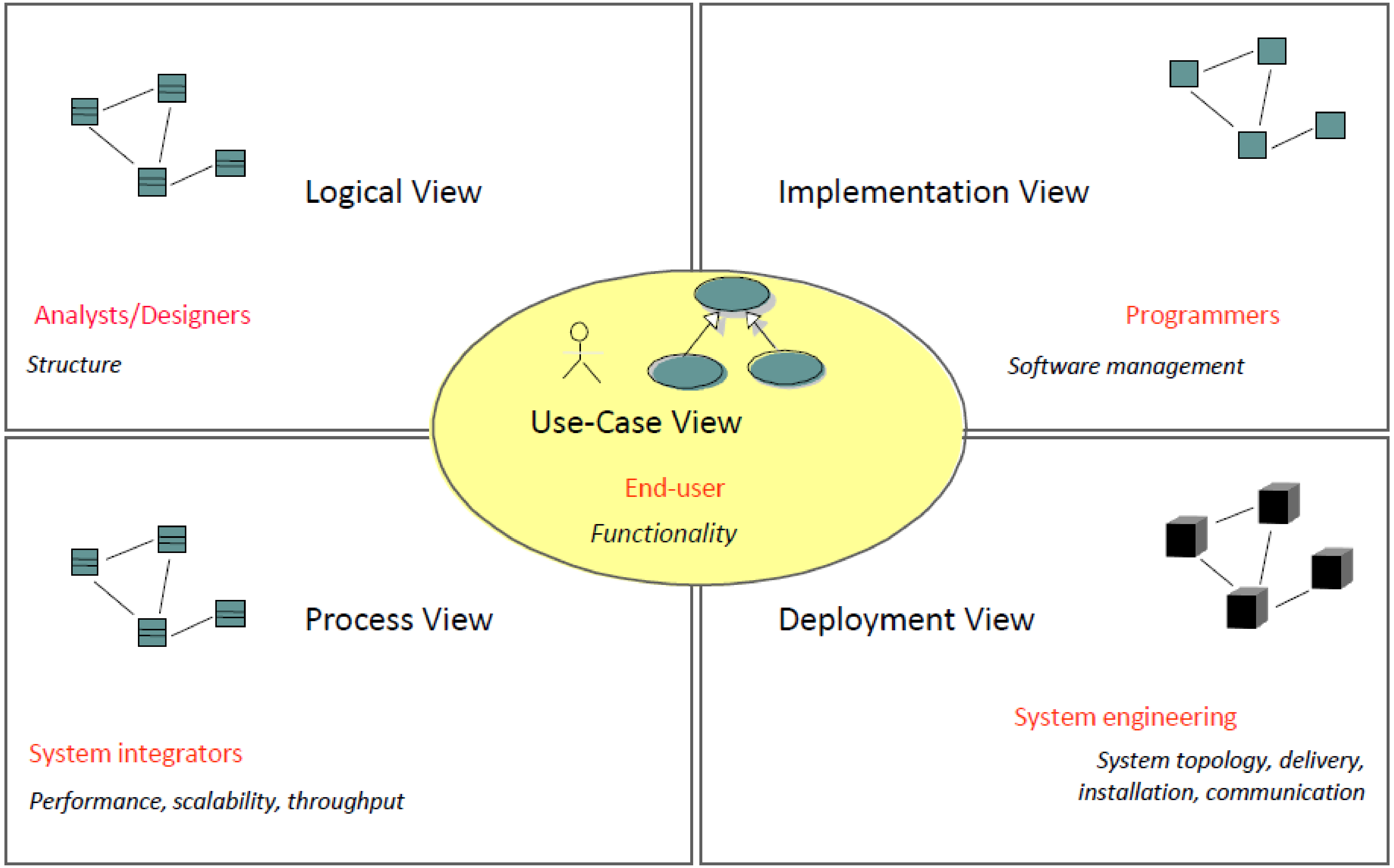
1. **Why Unified Modeling Language (UML)?**
   - Visual way to design and understand systems
   - Easy to share and communicate ideas
   - Works with any programming language
   - Supports Object-Oriented development
   - Helps model complex systems clearly
   - Uses best practices in software engineering

I. Introduction to UML (cont)



UML 2.2: Consist of 14 types of diagram

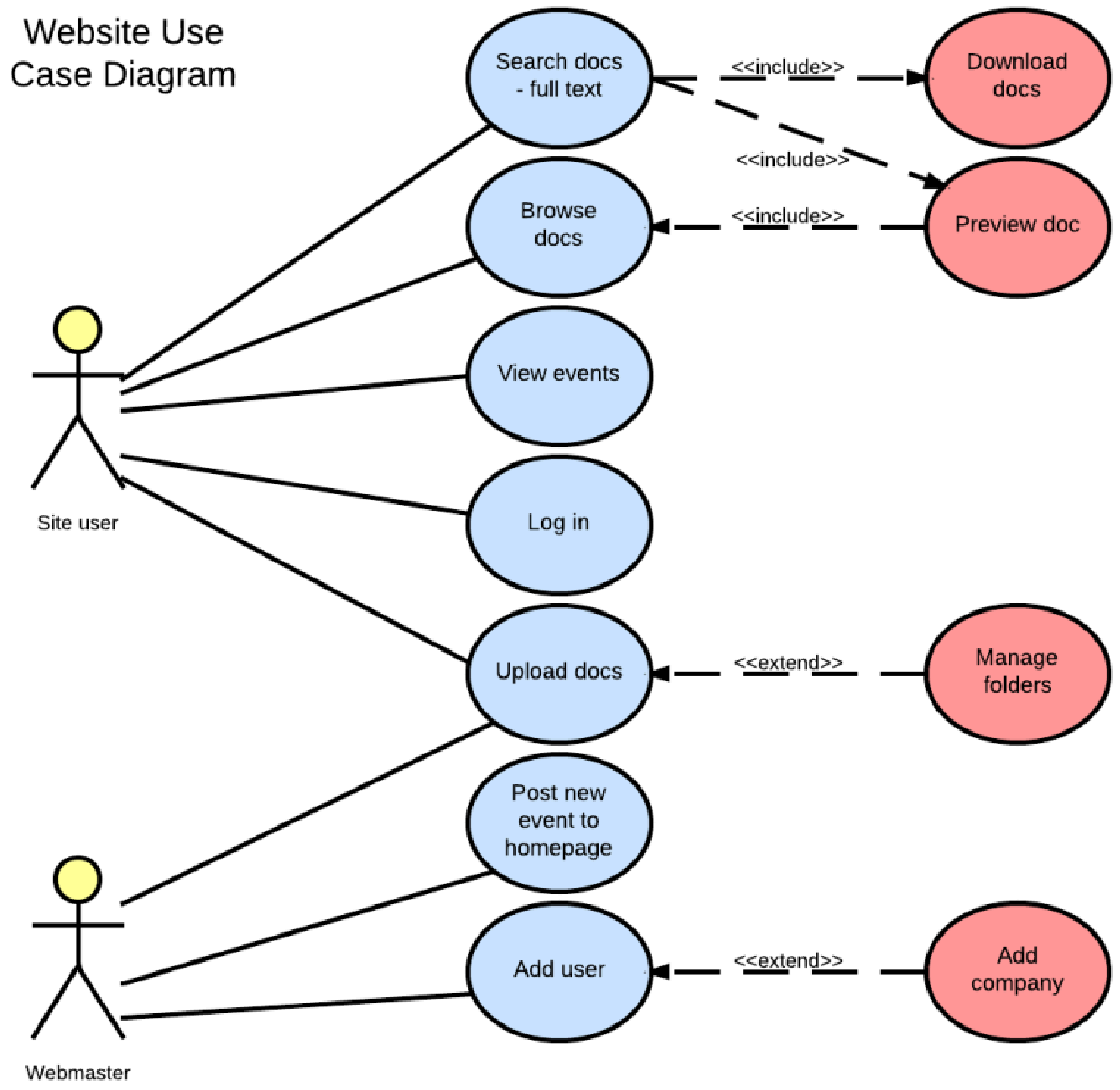
I. Introduction to UML (cont)



Different diagrams of system for different people

II. Use Case Diagram

1. **What is use case diagram?**
   - Use-case diagrams illustrate and define the context and requirements of either an entire system or the important parts of the system
   - You can create one or more to represent to model the components of the system.



Example: Use case Model

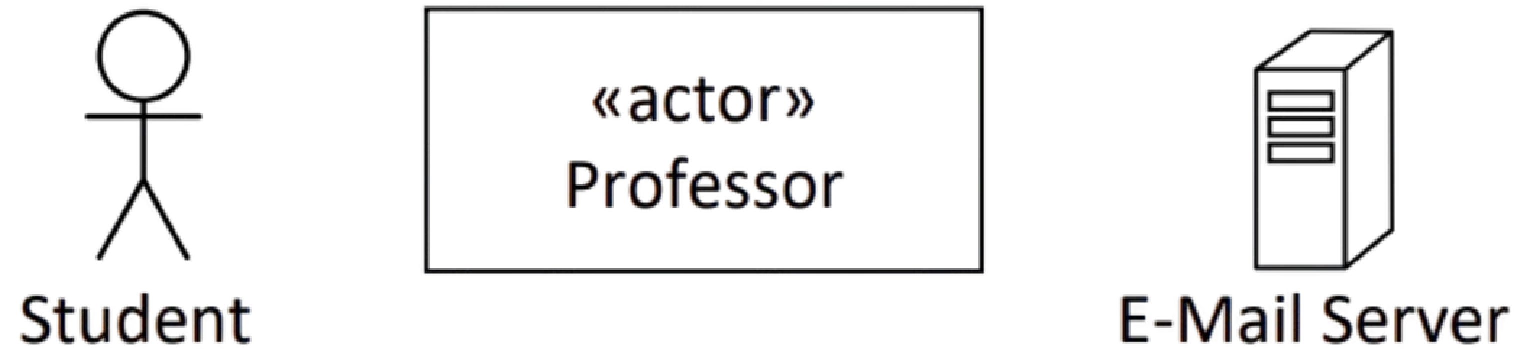
II. Use Case Model (Cont)

2. **When we create the use case diagram?**
   - Create in the early phase of the project
   - Refer to them throughout the development process
3. **How it helpful?**
   - Before start: to model a business so that all participants in the project share an understanding
   - While gathering requirement: to capture the system requirements and to present to others what the system should do.
   - During the analysis and design phases: to identify the classes that the system requires.
   - During the testing phase: to identify tests for the system.

III. Use Case Components

2. Actor

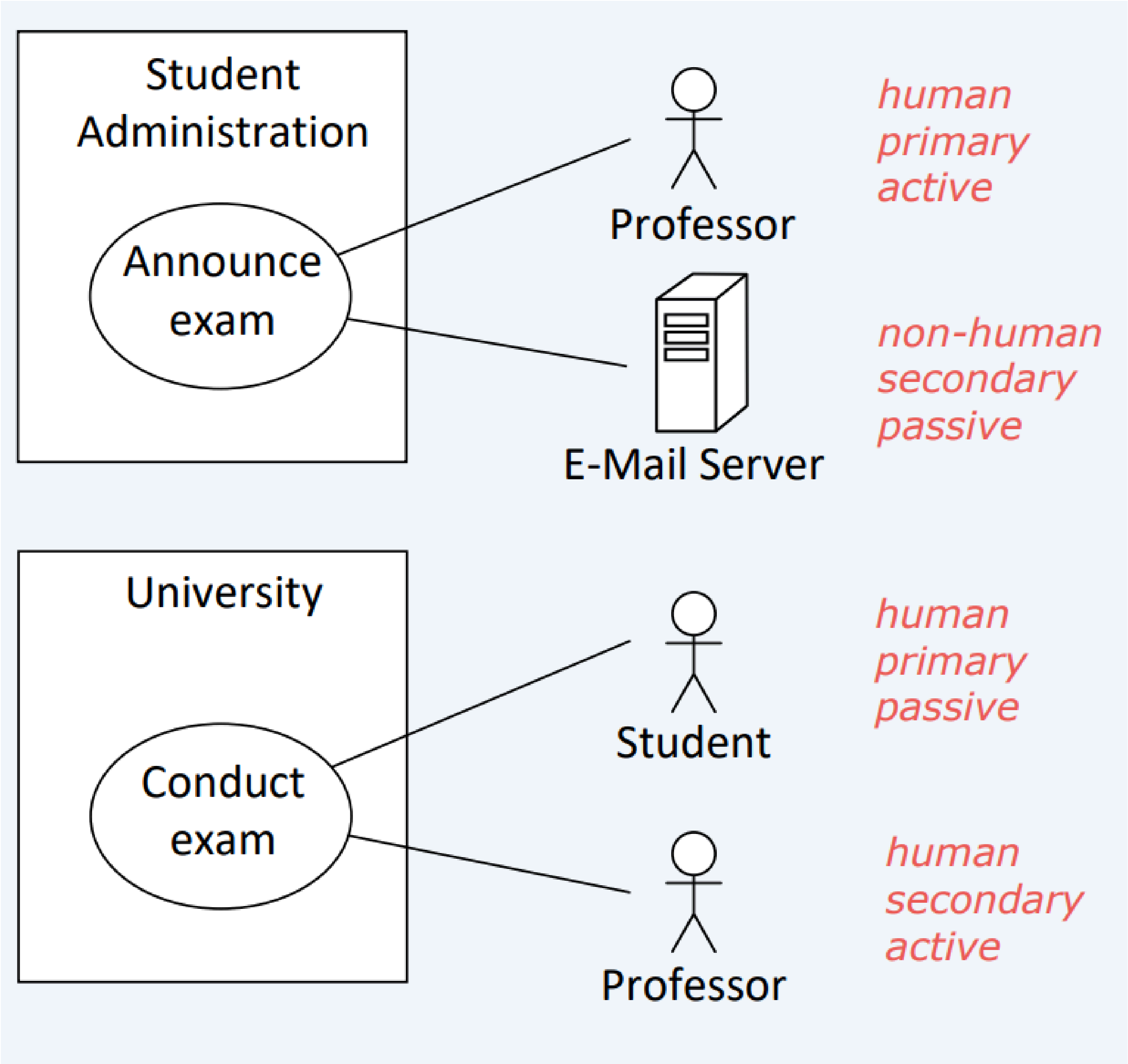
- Actors interact with the system:
  - By using the system
  - By being used by the system
- Actor is connected to the system by **associations**
- Actor are clearly outside the system boundaries



III. Use Case Components

2. Actor (Classification)

- Human
  - Student, Professor
- Non Human
  - Email-server
- Primary: main beneficiary of the application
- Secondary: necessary for the system to function
- Active: initiates use cases itself
- Passive: Does not initiate any use cases itself



III. Use Case Components (Cont)

2. Use Case

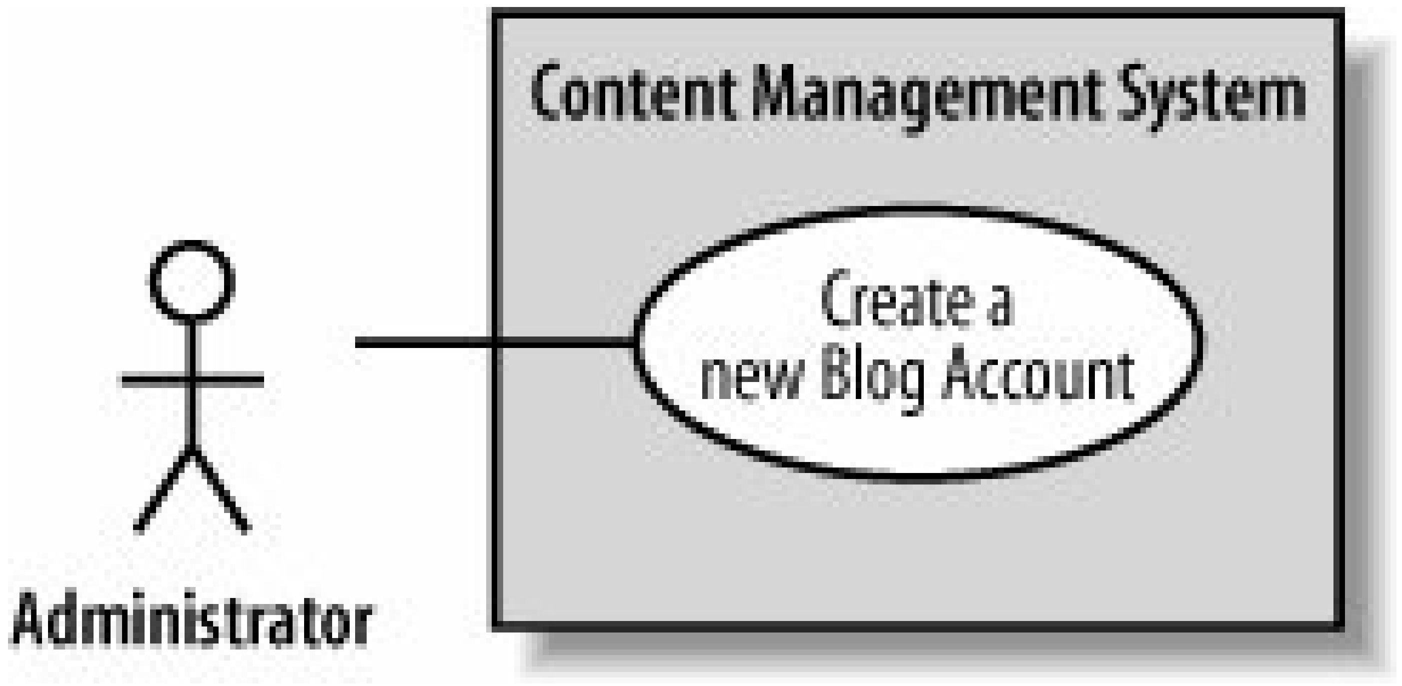
- Represents an interaction or task the system performs.
- Describes a specific job or function.
- Notation: Oval shape with the use case name inside.

Use case

III. Use Case Components (Cont)

3. System Boundary Boxes

- What is being described is called system. Ex: **Content Management System**
- Separates actors (external) from use cases (internal).
- Notation: Rectangle/box around use cases.
- Good practice to

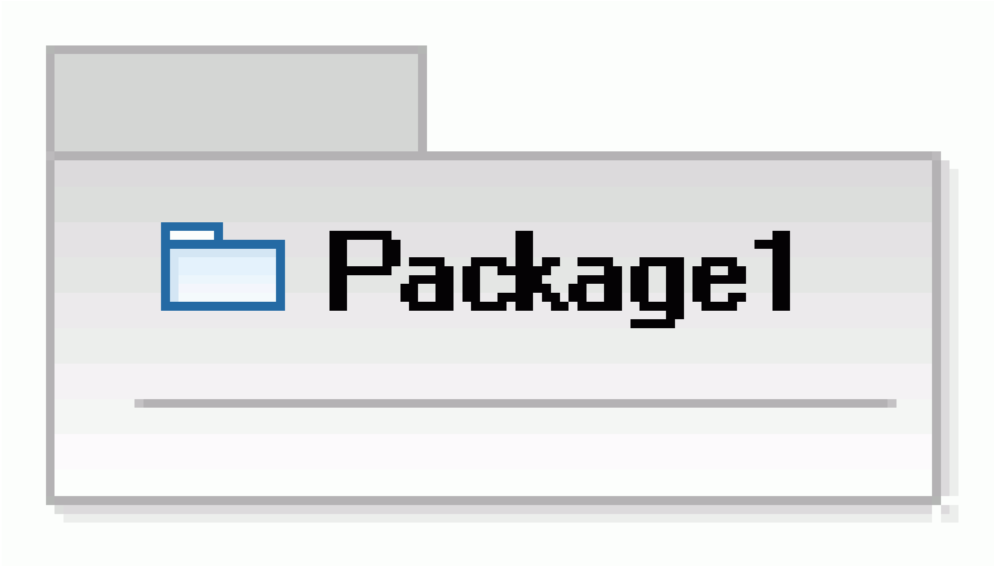


system Boundary Boxes

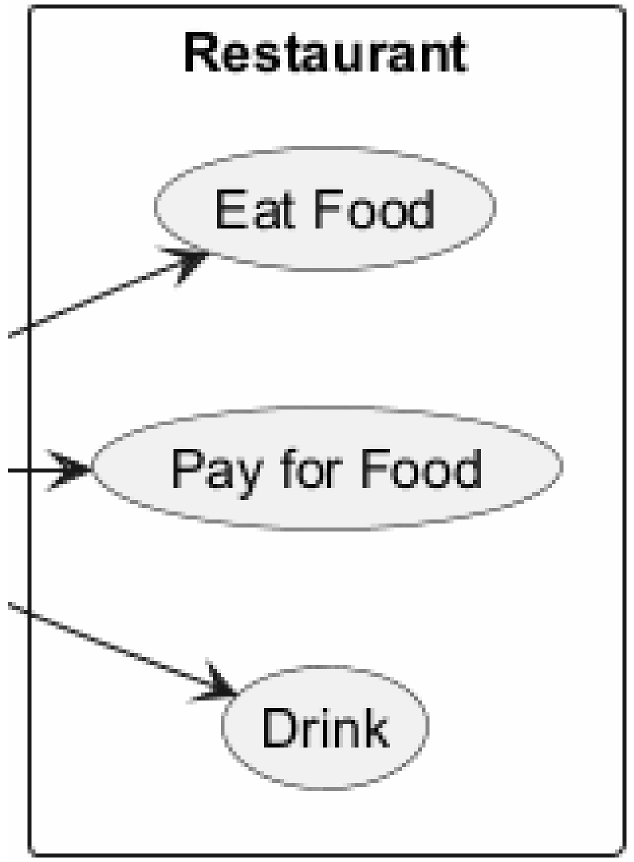
III. Use Case Components (Cont)

4. Package

- Groups related actors or use cases.
- Helps organize complex systems.
- Notation: Rectangle with a small tab or a simple rectangle.



Package



IV. Use Case Relationships

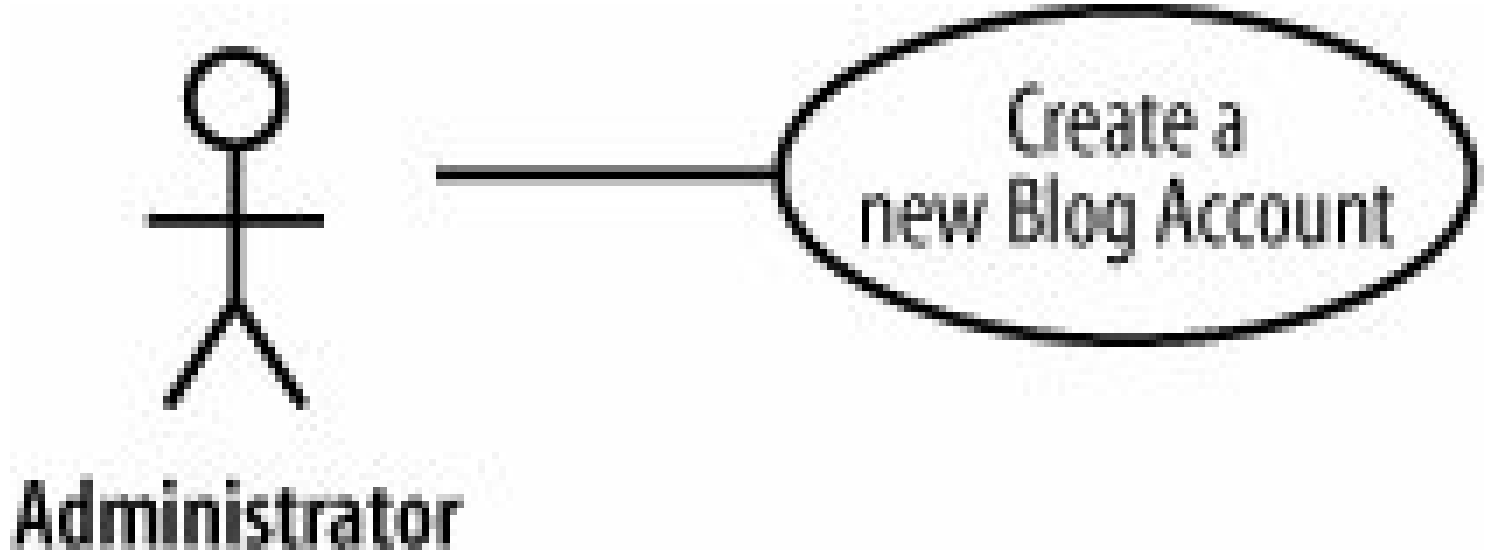
In UML a relationship is a connection between model elements.

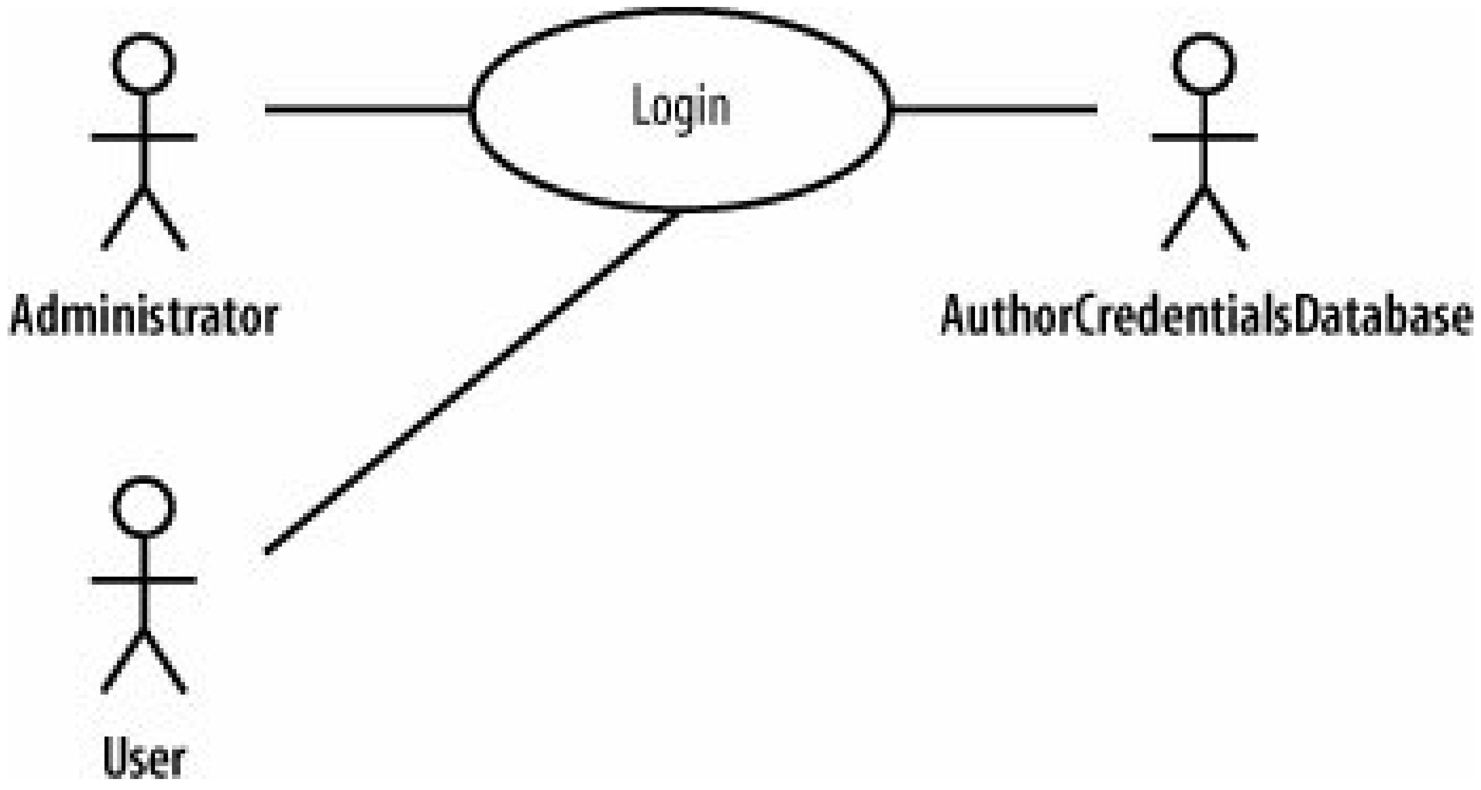
- Association Relationships (Communication Line)
- Generalization Relationships
- Include relationships
- Extend relationships

IV. Use Case Relationships (Cont)

1. Association Relationships (Communication Line)

- Connects an actor and a use case.
- Shows the actor participates in that use case..



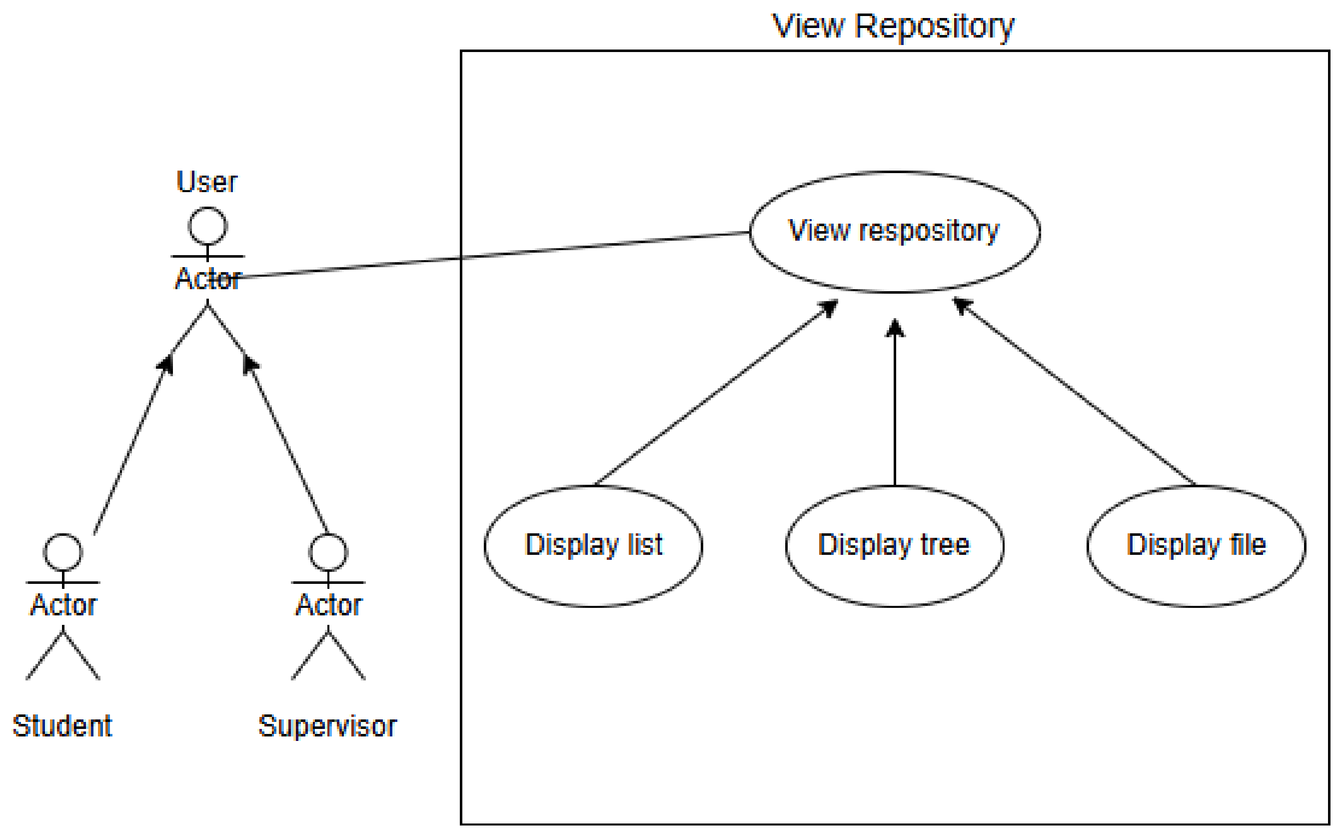


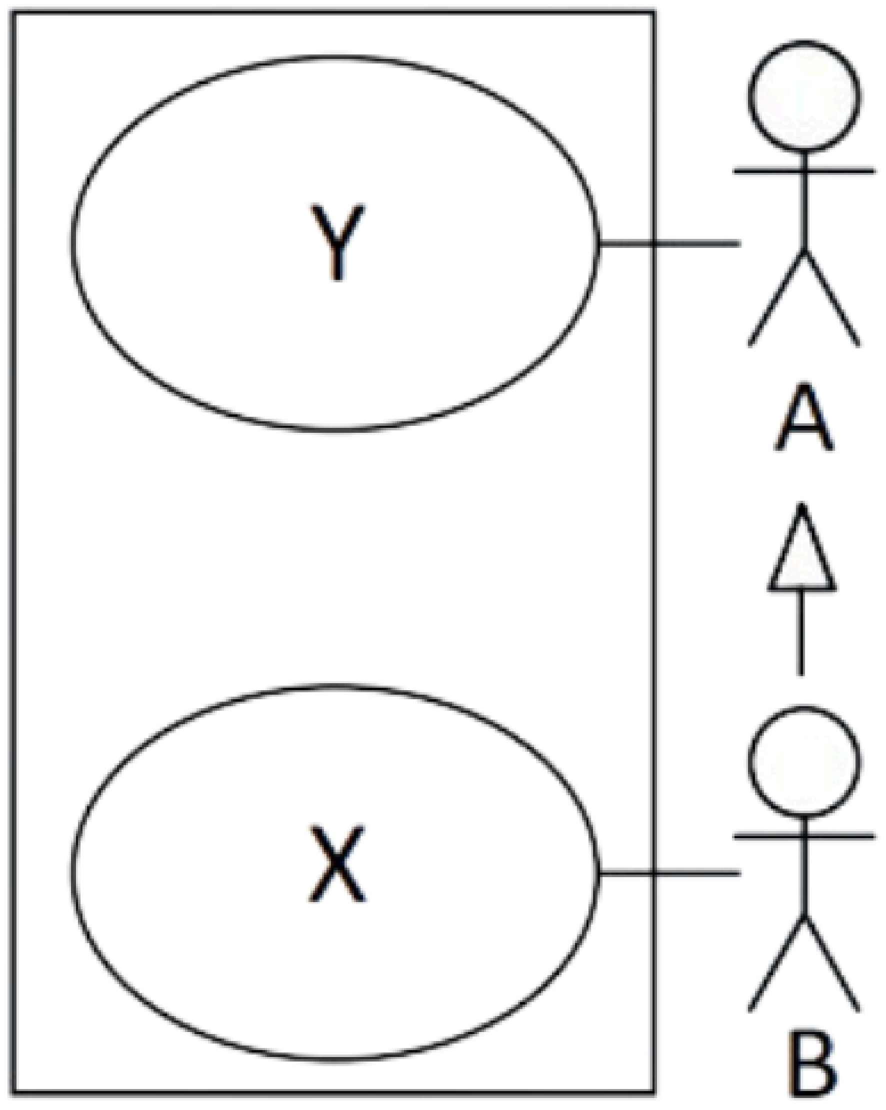
Association Relationship

IV. Use Case Relationships (Cont)

2. Generalization Relationships

- A use case or actor can inherit from another.
- The child can do everything the parent can do.
- Notation: Generalization arrow.

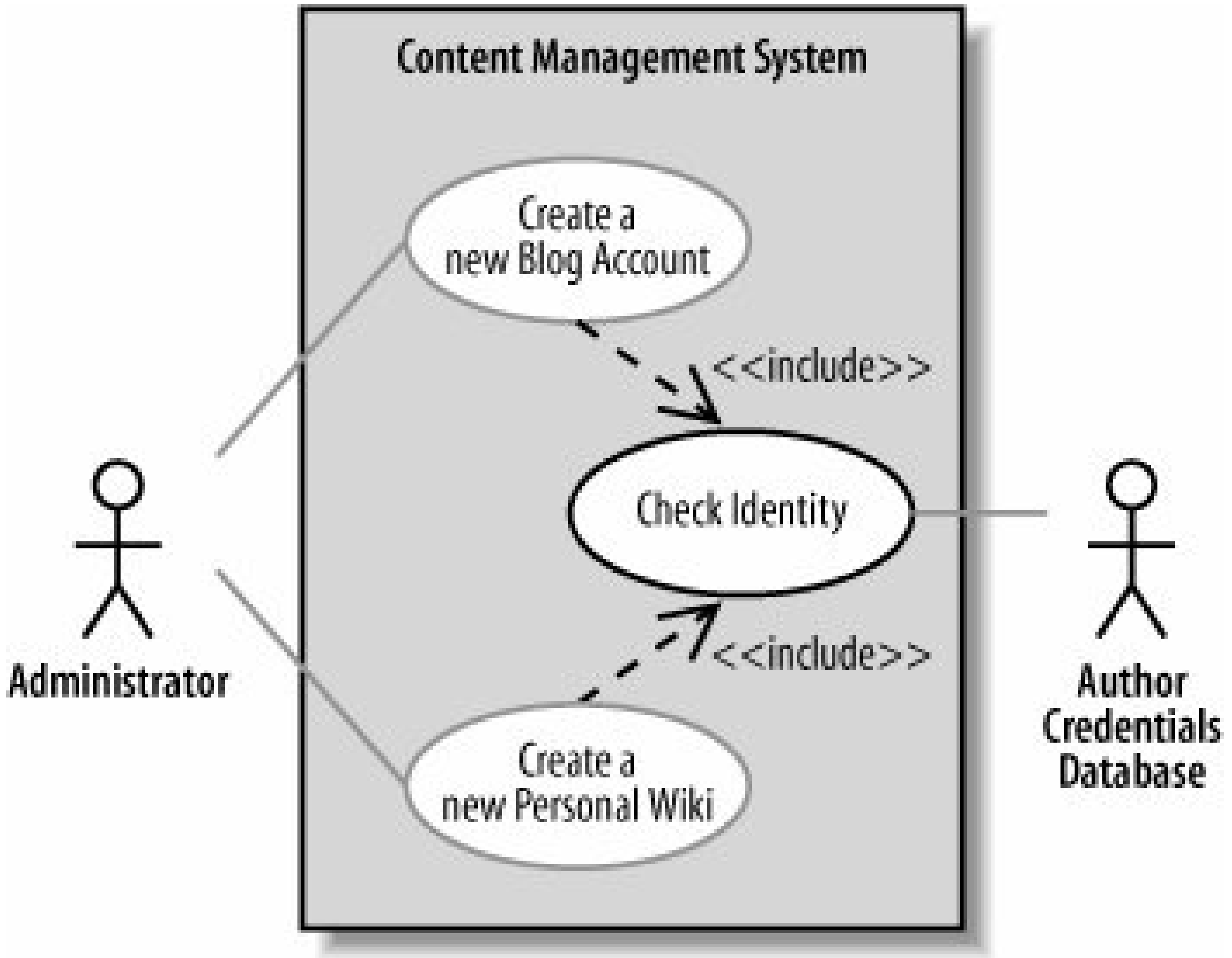




IV. Use Case Relationships (Cont)

3. Include Relationships

- Represents a use case reused by other use cases.
- Helps avoid repeating common functions.
- Notation: Dotted arrow labeled **<<include>>**.

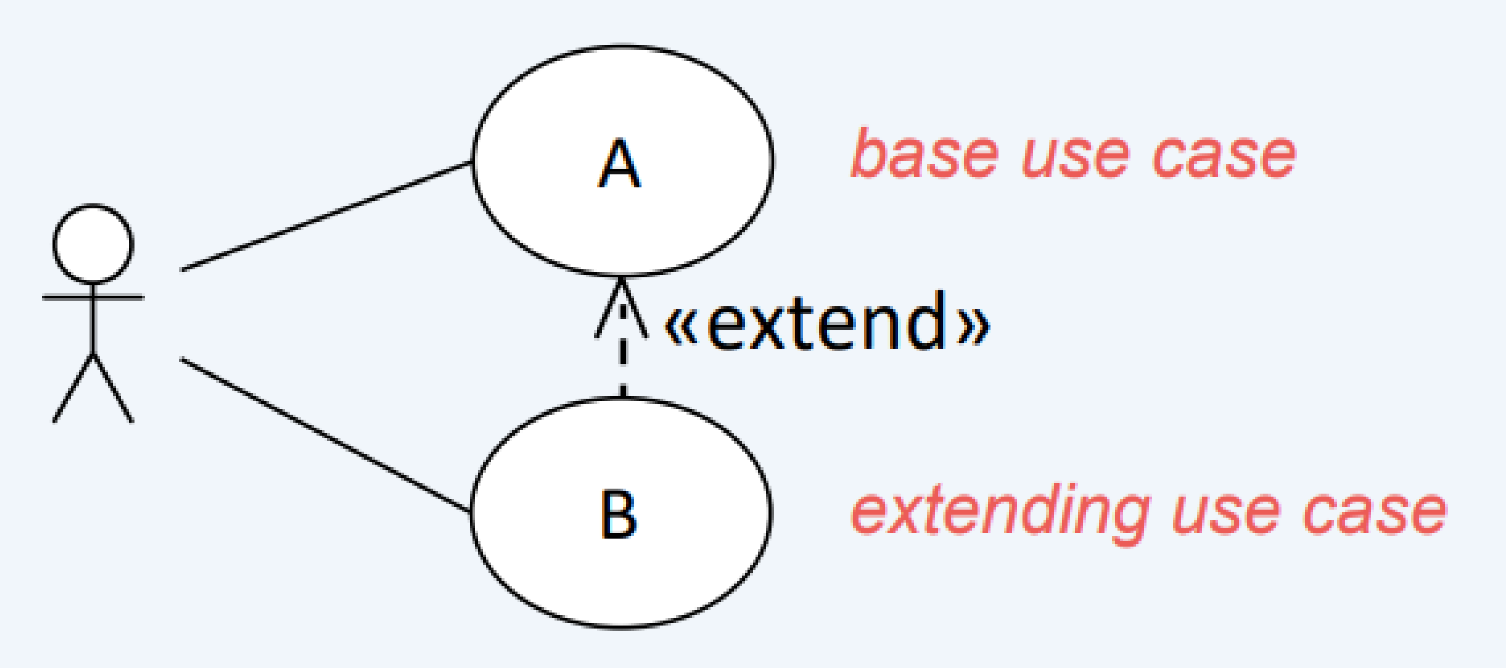


Include Relationship

IV. Use Case Relationships (Cont)

4. Extend Relationships

- Represents optional or additional behavior.
- Used when a function happens only under certain conditions.
- A and B can also be executed separately
- Notation: Dotted arrow labeled **<<extend>>.**



Research (Topic 1)

There some confusing error in building use case diagram

- Find as many as possible of those error confusing in use case diagram
- Find the error and the right version

Research (Topic 2)

There one more thing for use case diagram is called "Use case description"

- What is use case description?
- When we use it?
- Why we use in use case?
- etc

V. Tools

1. **StarUML**
   - Operation System: Window, Linux, MacOS
   - Link Download: https://staruml.io/download
2. **Draw.io**
   - Operation: Local (Window, Linux, MacOS), Online
   - Link Download: https://www.drawio.com/

VI. Case Study

1. Case Study A

- A bank customer can use the ATM to log into the system and perform check account balance, withdraw money, transfer fund and print receipt. The bank will govern the each of the actions made by customers.

2. Case Study B

- Students required to perform subjects registration for new semester offered by faculty. Student will be able to register the some of the subject if they pass the prerequisted subject. Instructor will approve the student registration.

VI. Case Study

3. Case Study C

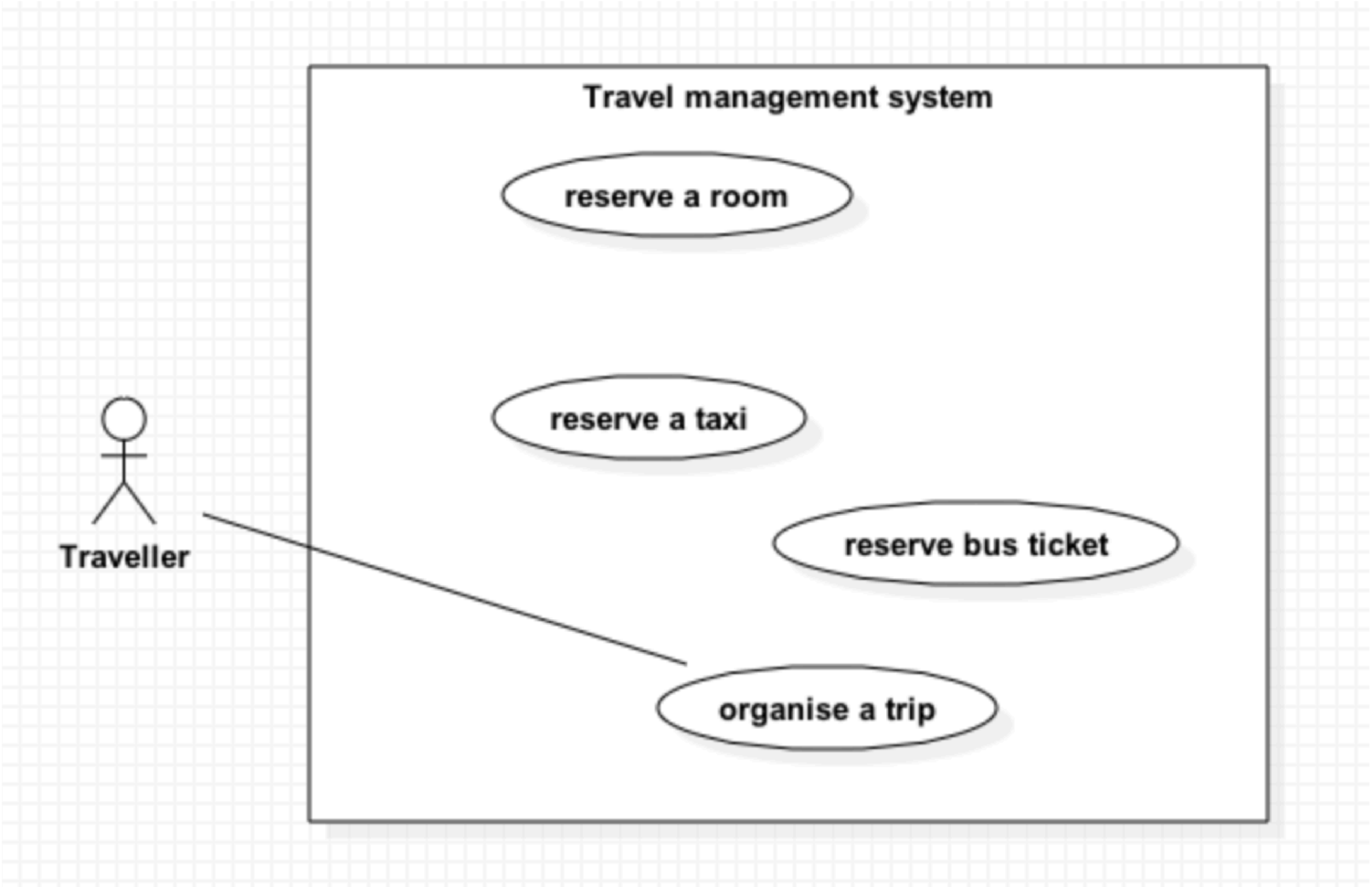
- A customer can use the Travel Reservation System to request an itinerary and confirm booking for travel services. The system allows the customer to book flights, reserve hotel rooms, and make payments. The staff prepares the itinerary, makes bookings, and checks payment status. The booking system manages booking information, while the credit card system verifies and processes payments. The system coordinates all these actions to ensure successful travel reservations.

VI. Case Study

4. **Case Study D: Identify the relationship between these use cases:**
    - The The travel agency has proposed a trip and travelers will stay in a hotel. Travelers have to book a taxi in order to get to the hotel.
    - Some travelers have demanded the travel agency to provide them the detail of the receipt. How do you put this new use case into your existing system?
    - Travelers can go to the travel destination either by bus or by plane. How do we model this?

VI. Case Study

4. Case Study D: Identify the relationship between these use cases:



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Thank You